FEATURE ARTICLE | MINING AND TECHNOLOGY | NOVEMBER 2026

Digging Smarter

How artificial intelligence is rewiring the African mining value chain, where the evidence is strong, where the hype runs ahead of it, and what the continent must build to lead rather than follow

Reading time: about 14 minutes

IN BRIEF

Artificial intelligence in mining has split into two stories. In some corners of the industry, such as driverless haul trucks and machine-learning-guided exploration, it has moved from pilot to production. In others, it is still a slide in a strategy deck. For Africa, which holds roughly 30 percent of the world's critical mineral reserves but remains thinly explored, the upside is large and the path is conditional.

This article argues that the winners will not be the mines with the most pilots. They will be the ones that fix four unglamorous foundations first: trustworthy data, reliable power and connectivity, a workforce able to run the systems, and governance that keeps the value of African data and African ore on the continent.

Why this, and why now

Mining is being squeezed from two directions. Demand for copper, cobalt, lithium, nickel and manganese is climbing with electrification and data-centre build-out, while the easy discoveries have been made. One of the best-funded AI exploration companies, KoBold Metals, argues that the industry now needs more than ten times the capital to make a discovery than it did thirty years ago. That is a company's claim rather than an audited statistic, but it captures a widely shared view: the next generation of orebodies is deeper, subtler and more expensive to find.

Africa sits at the centre of that squeeze. The continent is estimated to hold around 30 percent of global critical mineral reserves, and demand for several of those minerals is projected to rise sharply by 2030. A July 2026 Atlantic Council report on AI and African mining points out that colonial-era maps and infrastructure still hinder access to much of that resource, and argues that AI-enabled mapping and mining tools could shorten discovery, reduce drilling time and modernise operations.

The conversation has also matured. At Mining Indaba in Cape Town in February 2026, one of the featured sessions was titled "AI in Mining: Separating the Hype from Reality", and the agenda leaned on agentic AI, systems that do not merely analyse but decide and act. One speaker, a technology architect at Boston Consulting Group, noted that Africa has yet to scale autonomous ecosystems, which is precisely why the

chance to leapfrog is large. African Mining Week, held in Cape Town in October, gave AI its own panel. Attention is no longer the scarce resource. Evidence and execution are.

What follows walks the mining value chain from exploration to the mill, separating what is demonstrably working from what is promised, with a focus on the African context.

1. Exploration: finding what is left

Exploration is the most capital-intensive, highest-uncertainty stage of mining, and the one where AI's promise is most radical. The traditional process is sequential and slow: regional surveys, soil and rock sampling, geophysics, then drilling, with geologists integrating fragmented datasets by experience and judgement. Machine learning offers a different approach: combine decades of drill results, geochemistry, geophysics and satellite imagery into probabilistic models of where mineralisation is likely, then decide what data to collect next to reduce uncertainty fastest.

The clearest African example is KoBold Metals' Mingomba project in Zambia's Copperbelt. The company announced the discovery in 2024, describing copper grades above 5 percent against a global mine average of roughly 0.6 percent, and an expected output of at least 300,000 tonnes a year at full operation. KoBold's AI platform, built around a data layer called TerraShed and its Machine Prospector modelling tools, is designed to guide each exploration decision. The company also reports a striking result from northern Quebec in 2022, where AI-guided drilling intersected nickel-sulphide mineralisation in eight of ten holes, around ten times the typical industry hit rate. It now holds dozens of exploration projects globally and has raised roughly a billion dollars from investors.

Two cautions apply. First, these are company-reported results, and a discovery is not a mine. Industry commentary notes that KoBold has not previously built or operated one, so moving Mingomba from discovery to production is a different test from the one AI has passed so far. Second, AI does not remove geology risk. It shifts where the risk sits, from "where do we drill" to "is our data good enough to trust the model". Models are only as good as the historical data fed to them, and much African geoscience data is sparse, paper-based, or locked in legacy formats.

That last point is the opening for African institutions. National geological surveys that digitise, clean and responsibly open up their archives are effectively building the raw material for AI exploration. Countries that do this well will attract more and better exploration capital per dollar of promotion spent, because explorers can screen ground faster.

2. Autonomous operations: the most proven use of AI

If exploration is where AI is most exciting, autonomous haulage is where it is most demonstrably commercial. In April 2026 Komatsu announced that it had commissioned its 1,000th autonomous ultra-class haul truck, the first manufacturer to reach that scale, with the milestone also extending autonomy into gold mining. Chinese supplier EACON says it has delivered more than 3,000 autonomous mining trucks across upwards of 40 projects, with over 1,500 of them battery-electric. Hitachi and Pronto have partnered on retrofit autonomy for mixed fleets, and cabless, purpose-built electric haulers are arriving. These are largely vendor-reported figures, but the direction is consistent across sources: autonomy is moving from pilot to fleet-scale deployment.

The business case rests on three things. Trucks that do not need shift changes, breaks or rest run more hours. Consistent driving reduces tyre and mechanical wear. And removing people from the pit floor, next to highwalls, blasting areas and dump points, removes a category of exposure altogether. Published payback periods and cost-per-tonne improvements vary widely between sources and between mines, and trade-press claims should be treated with caution. A sober reading is that large open-pit operations with long mine lives and tight haul-road discipline capture the clearest gains, while smaller operations face longer paybacks.

The more interesting development for Africa is the move down-market. Bolt-on autonomy kits and interoperability standards are making autonomy viable for mid-tier fleets, which describes much of the continent's mining base. Consultants now talk about "autonomy-native" mine design, with ramps, benches and haul roads engineered for sensors and software rather than retrofitted for them. For an African miner planning a new pit rather than converting an old one, that is a genuine leapfrog option, provided the supporting layer exists: reliable site connectivity, stable power and skilled remote-operations staff.

3. Safety at depth: South Africa's distinctive case

South Africa operates some of the deepest mines on earth. Mponeng, near Carletonville, reaches roughly four kilometres below surface. SRK Consulting's rock-engineering team has long noted that the country reserves the term "deep level" for mines beyond about 2,000 metres, a depth at which much of the world's mining would already be called ultra-deep. At those depths the dominant hazards are seismicity, rockfalls, heat and the sheer complexity of coordinating people and machines in narrow, shallow-dipping reefs, where moving workers away from the face is harder than in massive orebodies.

Digital and AI tools are being applied in several ways. At Gold Fields' South Deep mine, a highly mechanised operation nearly three kilometres down, the company has expanded automation and tele-remote equipment in selected sections and rolled out a monitoring and analytics platform, including a central system staff describe as the mine's digital "brain". The CSIR, working with the Mandela Mining Precinct and the Minerals Council South Africa, has developed a digital-twin risk tool for trackless mobile machinery to help prevent vehicle collisions. Vision-based AI is being layered on top of proximity detection, which tells you something is near but not why it is dangerous. Impala Platinum's chief executive has said the company plans to accelerate AI pilots at its Rustenburg operations specifically to reduce human error in safety processes.

It is tempting to go further and promise that AI will predict rockbursts hours or days in advance. Researchers are working on exactly this, using machine learning on dense seismic monitoring data, and the CSIR and others are advancing the underlying sensing. But reliable short-term prediction of individual seismic events remains an unsolved scientific problem. The credible near-term value is narrower and still important: better hazard mapping, earlier flagging of high-risk areas, and more consistent re-entry and exclusion decisions. Mining leaders and regulators should hold vendors to that standard of honesty, because overpromising in safety erodes the trust that adoption depends on.

4. Plant, planning and maintenance: quiet wins

Away from the headlines, some of the most reliable returns come from applying machine learning to the data mines already collect. Predictive maintenance on mills, crushers, pumps and fleets reduces unplanned

downtime. Short-interval production control and AI-assisted scheduling help close the gap between the plan and what actually gets mined. Sensor-based ore sorting and process optimisation can lift recovery and reduce energy and water use per tonne. None of this is cinematic, and much of it can start with data historians and maintenance logs that a mine already owns.

For African operations, the maintenance and planning layer is arguably the best place to begin. It requires less capital than autonomy, carries lower safety risk if a model is wrong, and builds the data discipline and internal skills that every later project will need.

5. Governance and licensing: AI beyond the mine gate

AI's role is not limited to the pit and the plant. Reporting in the Mail & Guardian in February 2026 noted that South Africa loses an estimated R70 billion a year to unregulated mining, and linked that to the case for a modern digital mining cadastre: a transparent, machine-readable record of licences and rights that can streamline approvals and reduce the opportunities for corruption. The same coverage noted a pledged one-billion-dollar UAE fund for AI-related development in Africa. Satellite imagery and computer vision can also help regulators detect illegal or unlicensed activity and monitor tailings, land disturbance and rehabilitation.

This is the area where governments, not companies, hold the lever, and where relatively modest digital investment can unlock outsized revenue and investor confidence.

Where the evidence stands

Stage	Maturity in 2026	Strength of evidence	Main constraint in Africa
AI-guided exploration	Early commercial	Striking but largely company-reported; discovery-to-mine conversion unproven	Sparse, legacy and paper-based geoscience data
Autonomous haulage	Fleet-scale in open pit	Strong: thousands of trucks, multiple vendors	Connectivity, power, remote-operations skills, capital
Safety perception and digital twins	Pilot to early rollout	Promising; independent outcome data still thin	Integration with legacy equipment and labour relations
Predictive maintenance and planning	Mature and widely applicable	Strong and low-risk	Data quality, change management
Licensing and regulation	Early	Compelling logic; few completed implementations	Institutional capacity and political will

Assessment by the author, drawing on the sources listed at the end of this article.

The hard parts nobody can skip

Data readiness

Every AI use case above is a data use case. Mines that cannot say where their data lives, who owns it and whether it is trustworthy will not benefit from the most sophisticated model. The unglamorous work of instrumenting equipment, standardising formats and building data governance is the true start line.

Power and connectivity

Autonomy and remote operations depend on stable, low-latency networks and reliable electricity. In many African mining regions, both are intermittent. This is also why the shift toward battery-electric fleets and on-site renewables matters: the same investment can address cost, emissions and the energy foundation that digital systems need.

Skills and the social licence

Autonomy changes jobs more than it eliminates them, though some roles do shrink. Operators become remote controllers, technicians become data-literate, and new roles appear in analytics and fleet supervision. In South Africa, where mining is deeply tied to employment, union relations and the legacy of events such as Marikana, introducing automation without credible reskilling and consultation is a business risk as much as an ethical one. Mines that co-design the transition with workers and communities will move faster and face fewer stoppages.

Data sovereignty and vendor lock-in

When a foreign AI company models African geology, who owns the resulting knowledge? Exploration data is a strategic national asset. Contracts that let companies train proprietary models on a country's geoscience data without any return to the host country risk replicating the extractive pattern in a new form. Governments should consider data-sharing terms, local-processing requirements and capacity-building commitments as standard conditions of access. Miners, for their part, should insist on open interfaces so they are not locked into one vendor's ecosystem.

Geopolitics

Analysts increasingly describe AI-enabled mining as another arena in the US-China competition for African minerals, with both sides deploying technology, finance and partnerships. African states hold real bargaining power here, but only if they negotiate as informed buyers with their own data and technical capacity, rather than as passive recipients.

A practical playbook

The right next move differs by who you are. A compact guide:

Who	Start here	Avoid
Mining companies	Fix data foundations. Begin with predictive maintenance and planning. Run autonomy and safety-AI pilots with clear baselines, independent measurement and a workforce transition plan.	Pilot sprawl with no path to scale; accepting vendor ROI claims unverified
Governments and regulators	Digitise geological archives. Build a transparent digital cadastre. Set data-sharing and local-capacity conditions for exploration access.	Treating AI as a technology purchase rather than a policy and data-governance task
Investors	Back enabling layers (data, connectivity, skills, local integrators) as well as headline exploration plays. Underwrite the discovery-to-production gap honestly.	Valuing AI exploration on discovery announcements alone
African technology builders	Build for African conditions: low-bandwidth, mixed-fleet, multilingual and mid-tier mine economics. Partner with surveys, universities and research precincts.	Copying tools designed for Pilbara-scale mines without adapting to local realities

Outlook: leapfrog or lag

The next three years will test whether AI in mining becomes a durable advantage or another wave of over-promised technology. Several markers are worth watching: whether discoveries credited to AI convert into producing mines; whether autonomy spreads to mid-sized and underground operations in Africa; whether independent, published safety outcomes back up vendor claims; and whether African governments secure real terms of trade on their data.

Africa has an unusual combination of assets: abundant and under-explored geology, deep operating experience in the world's most demanding underground conditions, research institutions such as the CSIR and the Mandela Mining Precinct, and a growing technology-builder community. The mining industry's technology debate has largely been conducted about Africa. The opportunity now is to have it conducted with Africa, and, wherever possible, led from it.

The decisive factor will not be who buys the most advanced software. It will be who builds the foundations beneath it: clean data, dependable power, capable people and sound governance. Those are slower to build than a pilot, and far harder to copy.

Sources and notes

Figures from vendors and company announcements are described as reported rather than independently verified. Sources consulted:

1. Atlantic Council, Africa Center, "Artificial intelligence and the future of African mining", July 2026.
2. Engineering News (Creamer Media), "Tech that thinks for itself: Agentic AI to take centre stage at Mining Indaba 2026", 1 October 2025.
3. Mail & Guardian, "Unearthing AI's real value in African mining", 11 February 2026; and reporting on AI funding and digital licensing, 19 February 2026.
4. African Mining Week 2026 programme coverage (Energy Capital & Power; tech.africa), 2026.
5. KoBold Metals company profiles and funding summaries (sp-edge.com; battery-tech.net; komo.ai), accessed November 2026.
6. iAfrica, "AI takes hold in African mining, from Zambia's copper fields to Mali's autonomous gold mine", 2026.
7. International Mining and Mining Magazine reporting on Komatsu's 1,000th autonomous ultra-class haul truck, 21 to 23 April 2026.
8. Skillings Mining Review, autonomous mining technology coverage including EACON and Hitachi/Pronto, 2026.
9. Proactive Investors, "AI is powering a new era of mineral discovery" (citing McKinsey estimates).
10. Gold Fields South Deep coverage (Skillings; Mandela Mining Precinct visit, February 2026).
11. CSIR, "Mine digitalisation and automation" (Trackless Mobile Machinery Digital Twin).
12. JSE Magazine, "Out of harm's way" (Impala Platinum AI safety pilots); Engineering News, "The future of mining safety involves transforming fragmented controls into intelligent systems", 30 January 2026.
13. Engineering News, "Technology imperative to improve operations, safety in deep mines" (SRK Consulting, 2018).